

EXP11&EXP12:ROUTING-CPT

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Course Code: BCSE308L

AIM1: To configure and implement OSPF routing protocol in Cisco Packet Tracer and verify network connectivity using routing tables and ping.

Procedure:

- 1. Created a network topology using routers, switches, and PCs in Cisco Packet Tracer.**
- 2. Assigned IP addresses to all router interfaces and PCs.**
- 3. Enabled interfaces using no shutdown command.**
- 4. Configured OSPF routing protocol using router ospf and network commands on all routers.**
- 5. Verified routing using show ip route and tested connectivity using ping.**

IP addresses feeding of each router:

R1

```
conf t
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.255 area 0
end
```

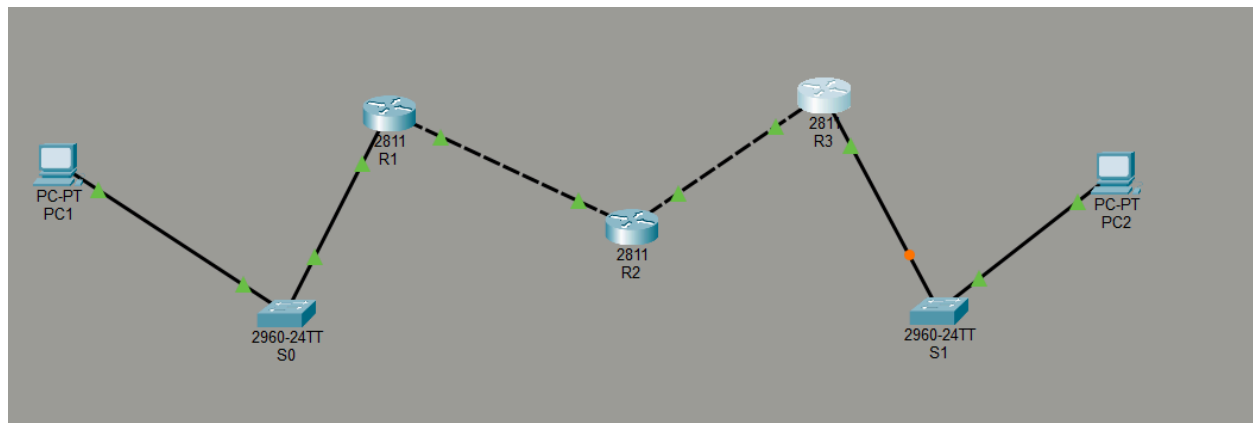
R2

```
conf t
router ospf 1
network 10.0.0.0 0.0.0.255 area 0
network 20.0.0.0 0.0.0.255 area 0
end
```

R3

```
conf t
router ospf 1
network 20.0.0.0 0.0.0.255 area 0
network 192.168.2.0 0.0.0.255 area 0
```

ROUTER DESIGN:



ROUTING TABLE OF ROUTER:

O INDICATES OSPF IS BEING LEARNED BY NETWORK

```
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.0 0.0.0.255 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

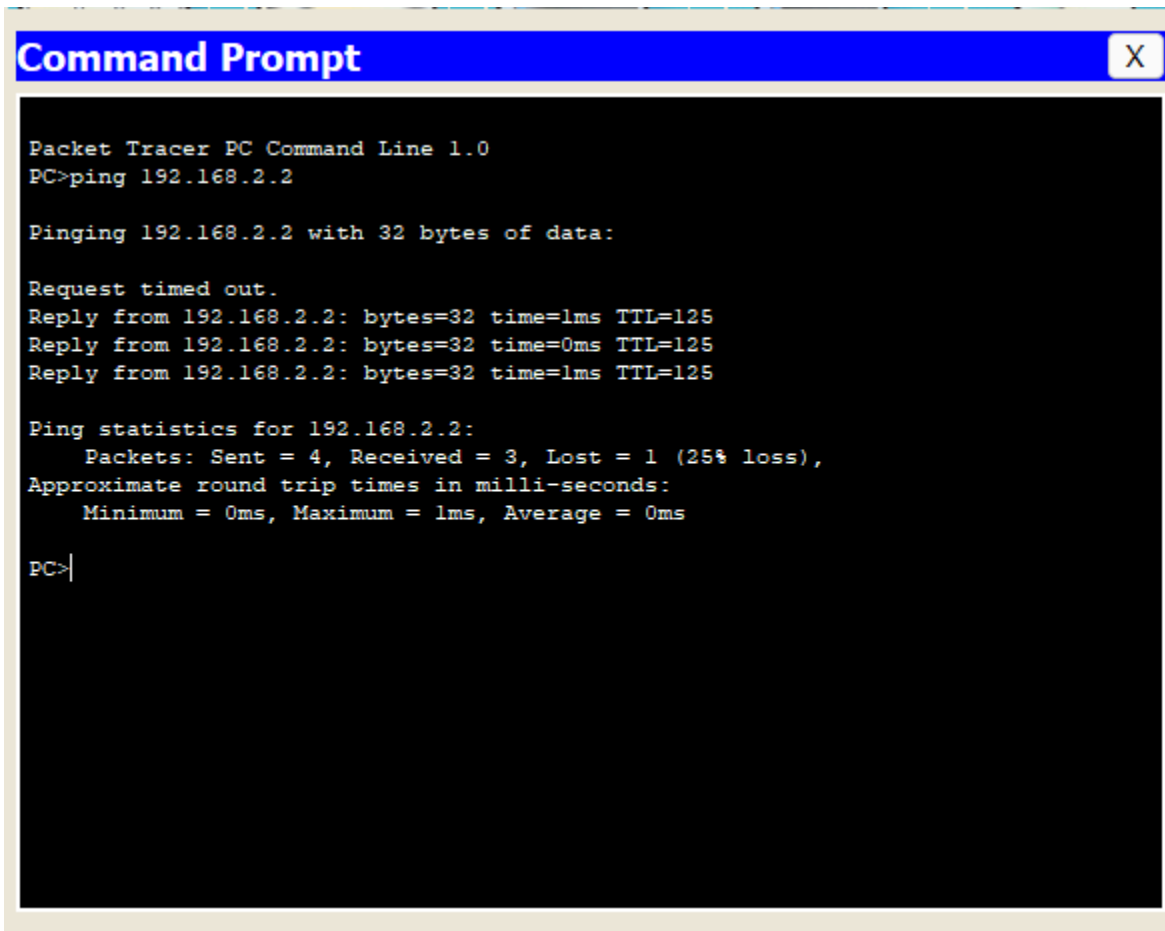
Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 1 subnets
C       10.0.0.0 is directly connected, FastEthernet0/1
    20.0.0.0/24 is subnetted, 1 subnets
O       20.0.0.0 [110/2] via 10.0.0.2, 00:08:38, FastEthernet0/1
C       192.168.1.0/24 is directly connected, FastEthernet0/0
O       192.168.2.0/24 [110/3] via 10.0.0.2, 00:08:38, FastEthernet0/1

Router#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
20.0.0.1       1     FULL/BDR        00:00:33    10.0.0.2       FastEthernet0/1
Router#
```

THIS IS THE COMMAND TERMINAL OF PC1 WHICH SHOWS THE TRANSFER OF MESSAGE



```
Command Prompt
X

Packet Tracer PC Command Line 1.0
PC>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time=1ms TTL=125
Reply from 192.168.2.2: bytes=32 time=0ms TTL=125
Reply from 192.168.2.2: bytes=32 time=1ms TTL=125

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC>|
```

AIM2: To configure and implement RIP routing protocol in Cisco Packet Tracer and verify network connectivity using routing tables and ping.

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set
```

IP CONFIGURE:

R1

```
conf t
router rip
version 2
network 192.168.1.0
network 10.0.0.0
End
```

R2

```
conf t
router rip
version 2
network 10.0.0.0
network 20.0.0.0
End
```

R3

```
conf t
router rip
version 2
network 20.0.0.0
network 192.168.2.0
End
```

After PING Command:

```
R 192.168.2.0
R 192.168.1.0
```

AIM3: To configure and implement EIGRP routing protocol in Cisco Packet Tracer and verify network connectivity using routing tables and ping.

IP config:

R1

```
enable
conf t
interface fa0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
interface fa0/1
ip address 10.0.0.1 255.255.255.0
no shutdown
end
```

R2

```
enable
conf t
interface fa0/0
ip address 10.0.0.2 255.255.255.0
no shutdown
interface fa0/1
ip address 20.0.0.1 255.255.255.0
no shutdown
End
```

R3

```
enable
conf t
interface fa0/0
ip address 20.0.0.2 255.255.255.0
no shutdown
interface fa0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
end
```

ROUTING TABLE:

```
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      10.0.0.0/24 is directly connected, FastEthernet0/1
L      10.0.0.1/32 is directly connected, FastEthernet0/1
      20.0.0.0/24 is subnetted, 1 subnets
D      20.0.0.0/24 [90/30720] via 10.0.0.2, 00:00:33, FastEthernet0/1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/24 is directly connected, FastEthernet0/0
L      192.168.1.1/32 is directly connected, FastEthernet0/0
D      192.168.2.0/24 [90/33280] via 10.0.0.2, 00:00:18, FastEthernet0/1
```

***The routing table shows routes learned using EIGRP, indicated by 'D'. The router successfully learns remote networks dynamically and forwards packets using the best path.**

COMMAND PROMPT OF PC1

```
Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```